IoT Door Lock System – Graph Plan

Predicates: DoorLocked, DoorUnlocked, SystemIdle, FaceDetected, Authorized, Unauthorized, AlarmOn
Initial State (S0): DoorUnlocked ∧ SystemIdle ∧ ¬AlarmOn ∧ ¬FaceDetected ∧ ¬Authorized ∧ ¬Unauthorized
Goal State: DoorLocked ∧ AlarmOn

Plan Layers:
S0: DoorUnlocked, SystemIdle, ¬AlarmOn, ¬FaceDetected
A0: DetectFace
S1: FaceDetected
A1: VerifyUnauthorized, VerifyAuthorized
S2: Unauthorized, DoorUnlocked
A2: LockDoor, TurnAlarmOn
S3: DoorLocked, AlarmOn

Mutex Examples:
State Mutex – DoorLocked ↔ DoorUnlocked, Authorized ↔ Unauthorized
Action Mutex – VerifyUnauthorized ↔ VerifyAuthorized, LockDoor ↔ UnlockDoor

IOT Door Lock System - SAT Plan:

Propositions: DoorLocked_t, DoorUnlocked_t, SystemIdle_t, FaceDetected_t, Authorized_t, Unauthorized_t, AlarmOn_t
Actions: DetectFace_t, VerifyUnauthorized_t, VerifyAuthorized_t, LockDoor_t, UnlockDoor_t, TurnAlarmOn_t, ResetSystem_t

Initial Clauses (t=0): DoorUnlocked_0, SystemIdle_0, ¬AlarmOn_0, ¬FaceDetected_0, ¬Authorized_0, ¬Unauthorized_0
Goal Clauses (t=3): DoorLocked_3, AlarmOn_3

$Function\ SATPlan\ (SmartDoorLock_Problem, Tmax)$
$for\ T\ =\ 0\ to\ Tmax\ do$
$\quad cnf, mapping\ \leftarrow\ Trans_to_SAT\ (SmartDoorLock_Problem, T\ )$
$\quad //\ Encode\ all\ CNF\ clauses:$
$\quad //-Initial\ state: DoorUnlocked_0, SystemIdle_0, \neg AlarmOn_0, \neg FaceDetected_0$
$\quad //-Goal\ state: DoorLocked_T\ \wedge\ AlarmOn_T$
$\quad //-Action\ preconditions, effects, frame\ axioms, and\ mutex\ constraints$
$\quad assignment\ \leftarrow\ SAT_Solver(\ cnf\ )$
$\quad if\ assignment\ \neq\ NULL\ then$
$\qquad return\ Extract_Solution(\ assignment, mapping\ )$
$end\ for$
$return\ failure$

SIH Write-Up
AI-Powered Railway Track Fittings Management System

Problem Statement
Indian Railways faced significant challenges in tracking and managing track fittings due to lack of identification systems for elastic rail clips, rail pads, liners, and sleepers and no integration with existing UDM and TMS portals and manual tracking leading to quality assessment and performance management issues and absence of mobile- based scanning capabilities for vendor lot numbers, supply dates, warranty periods, and inspection dates.

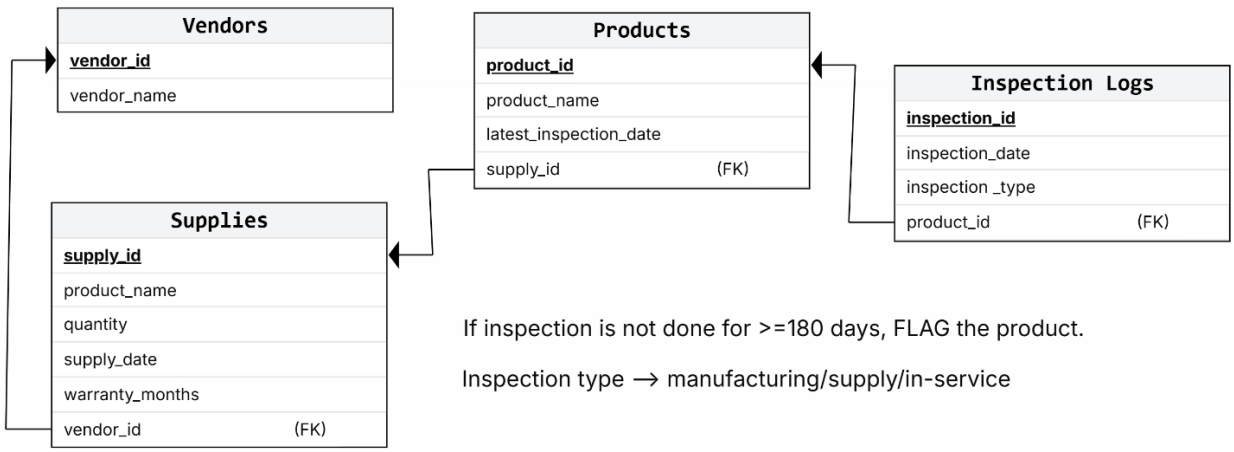
Currently, there is no unique identification mechanism for individual track fittings, and it is impossible to trace components from manufacturing to installation. There is also difficulty in tracking warranty periods and maintenance schedules, meaning challenges arise in recall management and quality control.

Indian Railways relies heavily on manual record-keeping and paperwork systems, which results in high error rates during data entry and retrieval processes. Inventory audits and inspections are extremely time-consuming, requiring significant manpower and often leading to operational inefficiencies. Moreover, the limited accessibility to historical data creates significant challenges in tracking component performance, warranty compliance, and maintenance scheduling across the vast railway network.

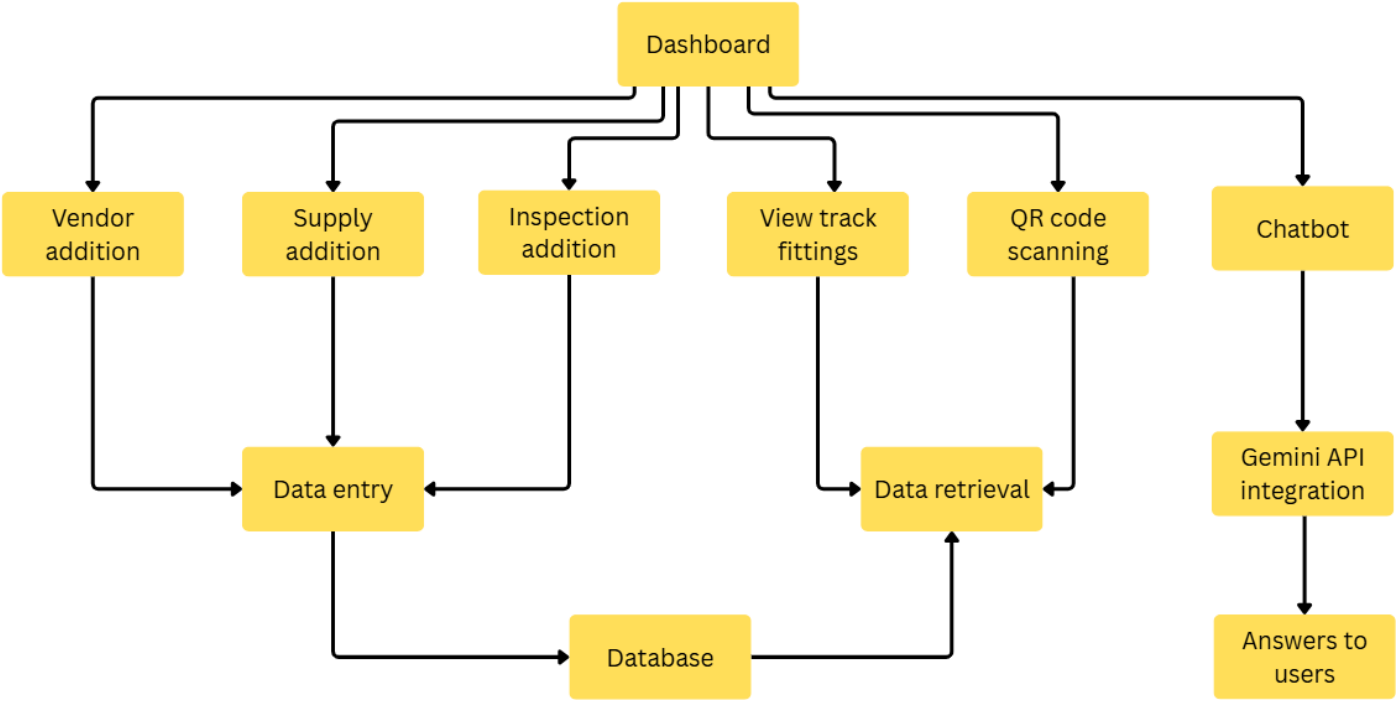
Objectives
- Develop a unified identification system for railway track fittings using laser-based QR code marking.
- Enable mobile-based scanning for real-time identification and data retrieval.
- Implement AI-powered querying for intelligent data analysis and reporting.
- Establish comprehensive quality monitoring from manufacturing to in-service performance tracking.
- Create 100% traceability for all track fittings from vendor to installation.
- Provide instant access to vendor information, supply dates, warranties, and inspection history.
- Reduce manual tracking efforts and improve data accuracy.
- Develop a user-friendly interface for data entry and management and successfully implement LLM-powered query processing.

Technologies Used
Backend: Flask, SQLAlchemy
Frontend: HTML5, CSS3, JavaScript
Database: SQLite
Chatbot: Google Gemini API

Database Schema



Workflow Diagram



Conclusion

This project successfully demonstrates that AI-powered solutions can revolutionize traditional industrial processes, making them more efficient, intelligent, and user- friendly. The Railway Track Fittings Management System not only solves immediate operational challenges but also establishes a new paradigm for railway infrastructure management in India. By combining laser-based QR code marking, comprehensive database management, and cutting-edge AI technology, this system provides a scalable, reliable, and future-ready solution that can significantly enhance railway safety, efficiency, and maintenance practices.

Chatbot Write-Up
AI-Powered Railway Track Fittings Management System

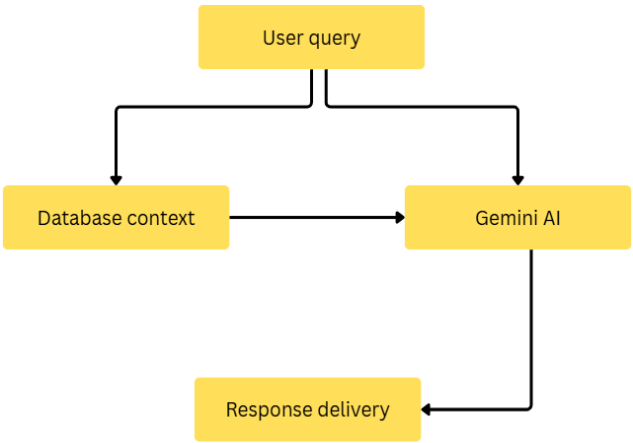
Problem Statement
The Railway Track Fittings Management System manages critical infrastructure components worth crores of rupees annually. Railway engineers, technicians, and procurement officers need instant access to complex data about vendors, supplies, products, and inspection histories. However, traditional query interfaces require users to navigate multiple screens and understand technical database structures.

Currently, railway engineers and technicians face significant challenges when trying to access critical track fitting information, as they must navigate complex web interfaces requiring multiple clicks for simple data retrieval tasks. The system demands technical knowledge to understand intricate database relationships, making it difficult for non-technical users to find relevant information efficiently. Time-consuming manual searches through extensive records further compound the problem, while the complete absence of natural language interfaces prevents users from asking contextual questions about component histories, maintenance schedules, or vendor performance in an intuitive manner.

Objectives
- Enable users to ask questions about track fittings data in plain English rather than using complex database queries.
- Create seamless integration between the AI chatbot and live system database for instant data retrieval.
- Train the AI to understand railway terminology, component relationships, and operational contexts.
- Understand and traverse complex relationships between vendors, supplies, products, and inspections.

Technologies Used
Core AI Framework: Google Gemini API (gemini-2.5-pro), Google Generative AI Library
Backend Integration Framework: Flask Framework, SQLAlchemy ORM
User Interface Technologies : HTML5, CSS3, JavaScript

Workflow Diagram



Conclusion
The AI chatbot integration has successfully transformed the Railway Track Fittings Management System from a traditional database application into an intelligent, conversational assistant. This integration demonstrates the transformative potential of AI in industrial applications, particularly in critical infrastructure sectors like railways. By combining Google's advanced Gemini AI with robust database architecture and user-centric design, the system now provides railway professionals with unprecedented access to complex component data through natural, intuitive conversations. This not only improves operational efficiency but also enhances safety, decision-making, and overall system reliability.

Protégé

Protégé is a free, open source ontology editor and a knowledge management system. The Protégé meta-tool was first built by Mark Musen in 1987 and has since been developed by a team at Stanford University. The software is the most popular and widely used ontology editor in the world.

1. Ontology Creation and Editing

- Supports OWL (Web Ontology Language) and RDF standards.
- Allows creation of classes, subclasses, and hierarchies.
- Enables definition of object properties (relationships) and data properties (attributes).
- Supports creation of individuals (instances) of classes.
- Provides annotation features for adding labels, comments, and metadata.

2. Reasoning and Inference

- Integrates with reasoners such as HermiT, Pellet, FaCT++, and ELK.
- Performs logical consistency checks and classification.
- Automatically infers subclass relationships and new facts based on axioms and rules.

3. Querying Capabilities

- Supports SPARQL queries for extracting and filtering ontology data.
- Allows users to query classes, individuals, and property relationships.

4. Visualization

- Provides class hierarchy and relationship visualization tools.
- Includes plugins such as OntoGraf and VOWL for graphical ontology exploration.
- Offers interactive network views to understand complex relationships.

5. Plugin and Extensibility Support

- Highly extensible through a modular plugin architecture.
- Supports plugins for rule-based reasoning (SWRL), visualization, ontology alignment, and data integration.
- Allows developers to create custom plugins using Java.

6. Ontology Import, Export, and Integration

- Supports multiple formats: RDF/XML, OWL, Turtle, JSON-LD, and others.
- Enables ontology import from external sources and repositories.
- Facilitates modular ontology development and reuse of existing vocabularies.

7. Collaborative Editing (WebProtégé)

- Web-based version for multi-user collaboration.
- Offers real-time editing, commenting, and discussion.
- Provides version control, access permissions, and activity tracking.

8. Rules and Constraints

- Supports SWRL (Semantic Web Rule Language) for defining logical rules.
- Enables domain-specific logic and advanced reasoning beyond standard OWL axioms.

9. Ontology Metrics and Validation

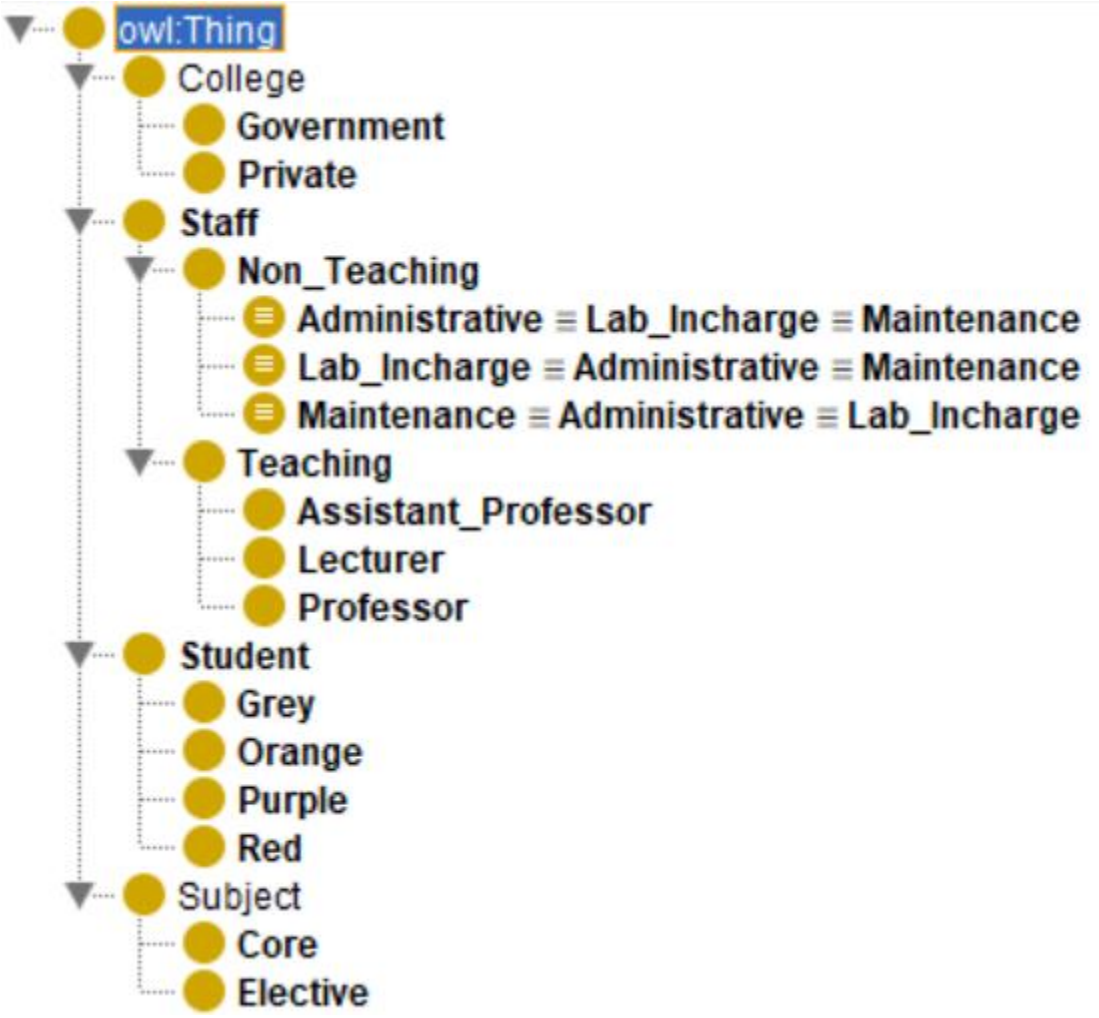
- Displays metrics such as number of classes, properties, and individuals.
- Checks logical consistency and detects modelling errors or missing elements.
- Provides validation reports for quality assurance.

10. Integration with Semantic Web Technologies

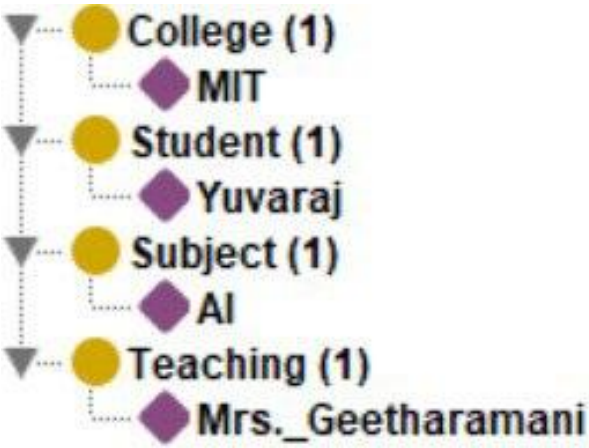
- Compatible with triple stores and semantic databases (e.g., Jena, GraphDB, Stardog).
- Supports linked data publishing and semantic reasoning.
- Can be integrated into AI systems and knowledge-based applications.

Ontology on College

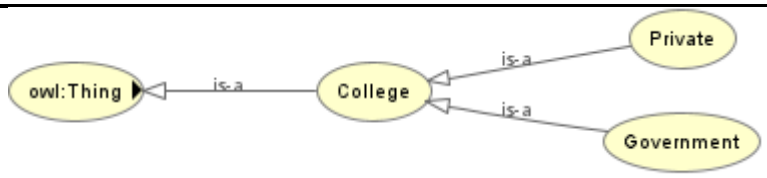
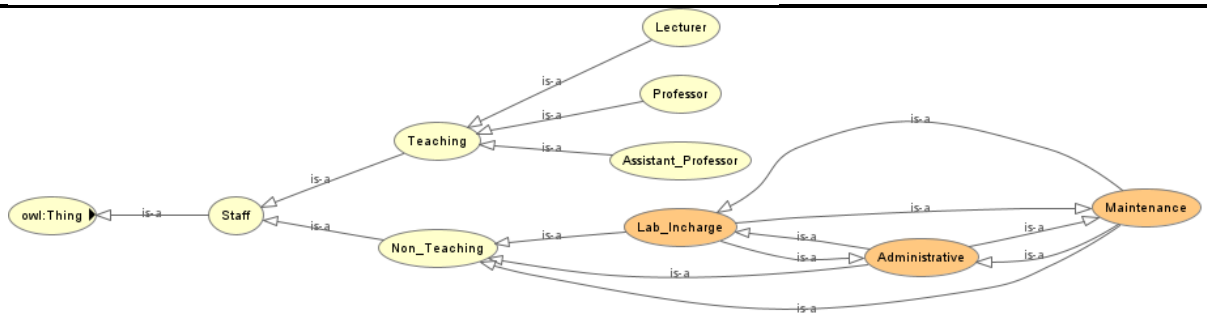
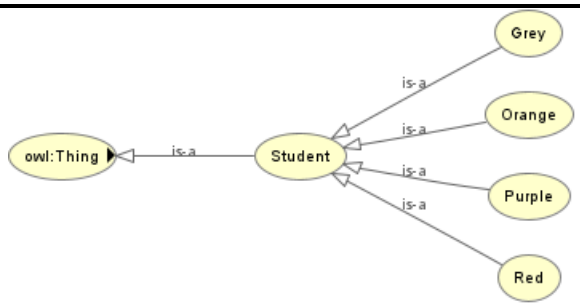
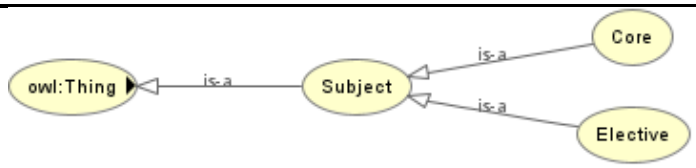
The hierarchical structure for the college ontology is given below.



Individual data has been added.



A visualisation of the constructed ontology is also generated.

Thing (Root)	
College	
Staff	
Student	
Subject